Competitive Deep Dive

hyper2kvm + hypersdk vs Microsoft SCVMM, Azure Migrate, virt-v2v & Others

Feature-by-feature, architecture, cost, and capability comparison across all major VM migration tools.
Why open-source wins for KVM/libvirt and KubeVirt targets.

Master Feature Matrix

All major VM migration tools compared head-to-head.

Capability	hyper2kvm + hypersdk	virt-v2v	Red Hat MTV	SCVMM	Azure Migrate	AWS MGN	Carbonite
VMware source	✓	✓	✓	✓	✓	✓	✓
Hyper-V source	✓	✓	Partial	✓	✓	✓	✓
Multi-cloud source	✓ (10)	✕	✕	✕	Azure	AWS	Limited
Target: KVM/libvirt	✓	✓	✕	✕	✕	✕	✕
Target: KubeVirt	✓	✕	✓	✕	✕	✕	✕
Offline guest fixes	✓ VMCraft	✓	Partial	✕	✕	✕	✕
Windows VirtIO 3-layer	✓	Basic	Basic	N/A	N/A	N/A	N/A
Web dashboard	✓ (20+ panels)	✕	✓	Portal	Portal	Console	✓
Guest agent insights	✓	✕	✕	✕	Partial	✕	✕
Security analysis	✓	✕	✕	✕	Partial	✕	✕
Air-gap / offline	✓	✓	Partial	✕	✕	✕	✕
Licensing	Proprietary (HyperSDK)	GPL	OCP req	$$$	Free tier	Free tier	$$$
Auto disk cleanup	✓	✕	✕	✕	✕	✕	✕
Credential persistence	✓	✕	✓	✓	✓	✓	✓

vs Microsoft SCVMM & Azure Migrate

Open-source migration vs Microsoft's proprietary, cloud-locked stack.

Microsoft Approach

```
VMware / Hyper-V
  ↓
Azure Migrate Appliance [requires internet]
  ↓
Azure cloud [locked-in, pay monthly]
  ↓
Azure Portal (management)
```

Lock-in: Azure AD required, outbound HTTPS, per-VM compute charges, data egress fees to leave.

hyper2kvm + hypersdk Approach

```
VMware / Hyper-V / 10 Clouds
  ↓
h2kvmctl + VMCraft [offline fixes]
  ↓
KVM/libvirt + KubeVirt [your servers]
  ↓
Web Dashboard (self-hosted)
```

Freedom: No cloud account, no internet needed, no per-VM charges, full infrastructure control.

Cost: 100-VM Migration (3 years)

Item	hyper2kvm + hypersdk	Microsoft Azure Migrate
License	**Contact for pricing***	Free (compute costs apply)
Target infra (3yr)	**Your existing servers**	**$150,000+ Azure compute**
Support	Lilo Technologies + commercial	$500+/hr Premier
Total	**Contact for pricing***	**$150,000+**

vs virt-v2v

Next-generation VMCraft engine vs legacy libguestfs appliance.

virt-v2v Architecture

```
Source → nbdkit → libguestfs [boots kernel!]
→ output disk → manual virsh define
```

~500MB deps, supermin required, no web UI, no KubeVirt, no multi-cloud, basic Windows

hyper2kvm Architecture

```
Source (10 clouds) → govc → qemu-nbd [no kernel]
→ VMCraft fixes → auto define + start + dashboard
```

~50MB install, any Linux distro, 20+ panel dashboard, KubeVirt deploy, 3-layer Windows

Aspect	hyper2kvm	virt-v2v
Engine	**VMCraft (qemu-nbd, no kernel)**	libguestfs (boots kernel appliance)
Install size	**~50MB pip**	**~500MB + kernel**
Cloud sources	**10 providers**	**VMware + Hyper-V only**
Windows VirtIO	**3-layer firstboot + SATA default**	**Basic injection**
KubeVirt deploy	✓	✕
Web dashboard	**20+ panels**	**None**
Progress streaming	**Real-time WebSocket**	**Batch stderr**
Auto VM start	**✓ (define + start)**	**Manual virsh define**
Cross-distro	**Fedora/RHEL/Ubuntu/Alma**	**RHEL/Fedora primary**

Windows Migration Deep Dive

The critical difference: hyper2kvm's 3-layer firstboot vs everyone else's "hope for the best."

Layer 1: rhsrvany Service

Windows service that runs VirtIO driver install at startup. Survives Safe Mode. Most reliable mechanism. Self-removing after completion.

Layer 2: RunOnce Registry Key

Writes to `HKLM\...\RunOnce`. Backup if service fails. Runs once at next login, auto-removes. Covers user-session scenarios.

Layer 3: Startup Folder

Script in All Users Startup folder. Final fallback — if service and registry both fail, this ensures firstboot runs when any user logs in.

SATA Boot Bus (Default)

Uses SATA as default disk bus. Windows has built-in SATA/AHCI drivers — VM **always boots** on first attempt. No VirtIO disk driver needed at boot time.

Why this matters: Microsoft's tools don't target KVM so they don't need VirtIO injection. virt-v2v does basic driver injection but uses VirtIO as boot bus — if drivers don't load, the VM blue-screens. hyper2kvm is the only tool with a production-grade, multi-fallback Windows migration path.

Windows OS Detection

hyper2kvm auto-detects Windows VMs even when the name doesn't contain "win" — by checking libvirt XML for `<hyperv>` features, `localtime` clock offset, and `<description>` tags. This powers correct driver injection, boot bus selection, and OS-specific dashboard badges.

hypersdk — The Export Engine

10 cloud providers, 238+ API endpoints, parallel export, carbon-aware scheduling.

10 Cloud Providers

VMware vSphere, Hyper-V, AWS EC2, Azure, GCP, Oracle Cloud, OpenStack, Alibaba Cloud, Proxmox, and more. Unified API across all sources.

238+ REST API Endpoints

Full lifecycle management: jobs, schedules, webhooks, libvirt domains, KubeVirt VMs, VM introspection, disk management, credentials, capacity planning.

Carbon-Aware Scheduling

Delays backup jobs to low-carbon grid periods. 30-50% carbon reduction. ESG compliance reporting with zone-based intensity tracking.

Parallel VM Export

Export multiple VMs simultaneously with per-file progress streaming. Configurable parallelism (1-32 concurrent downloads).

Web Dashboard

React/TypeScript dashboard: VM management, KubeVirt, capacity gauges, top consumers, activity feed, storage cleanup, credential store.

Job Scheduling

Cron-based scheduled exports. Create, update, enable/disable, trigger manually. Webhook notifications on completion/failure.

No other migration tool offers all of this: hypersdk is a complete VM export platform — not just a conversion tool. Combined with hyper2kvm's conversion engine, it provides the full migration pipeline from any cloud to self-managed KVM infrastructure.

Web Dashboard Capabilities

20+ panels for VM lifecycle management — included free. No other open-source tool has this.

- ✓ Guest Agent OS Insights
- ✓ Security Exposure Analysis
- ✓ Smart Recommendations
- ✓ Connectivity Testing
- ✓ Process Viewer (guest-level)
- ✓ Storage Detail & I/O Stats
- ✓ Network Interface Detail
- ✓ VM Events Timeline
- ✓ Disk Hot-Plug Management
- ✓ CDROM Insert/Eject
- ✓ Cluster Capacity Gauges
- ✓ Top Resource Consumers
- ✓ Activity Feed
- ✓ VM Create / Clone / Resize
- ✓ VM Import from Disk
- ✓ Snapshot Management
- ✓ Live Migration Monitoring
- ✓ Multi-Kubeconfig Mgmt
- ✓ Domain XML Viewer
- ✓ Auto Disk Cleanup
- ✓ Credential Persistence
- ✓ Batch Migration Queue
- ✓ Real-Time Progress
- ✓ HTTPS by Default

Self-Hosted & Air-Gap Ready

Dashboard at `https://host:5070`. No cloud, no internet, no external deps. Auto-generated ECDSA P-256 TLS certificates. Perfect for classified environments.

75+ API Endpoints (hyper2kvm)

Full REST API with auto-generated docs at `/api/v1/docs`. Session auth, audit logging, webhook notifications, email alerts. Production-ready.

Migration Workflow

Microsoft: 8 steps. virt-v2v: 5 steps. hyper2kvm: 3 steps.

Microsoft (8 steps)

1. Create Azure account
2. Deploy appliance
3. Configure discovery
4. Run assessment
5. Begin replication
6. Test migration
7. Cutover
8. Decommission

Timeline: 2-6 weeks per batch

virt-v2v (5 steps)

1. Install libguestfs
2. Run virt-v2v command
3. Wait (no progress)
4. Manual virsh define
5. Manual virsh start

Timeline: hours. Manual post-work.

hyper2kvm (3 steps)

1. `h2kvmctl --config m.yaml`
2. Watch dashboard
3. Done (auto-defined + started)

Timeline: minutes. Fully automated.

The bottom line: If your target is Azure, use Azure Migrate. If your target is *your own KVM/libvirt or KubeVirt infrastructure*, hyper2kvm + hypersdk is the only production-grade migration platform that exists. 3 steps. Contact for pricing.* Air-gap capable. 20+ dashboard panels. 10 cloud sources. Done.